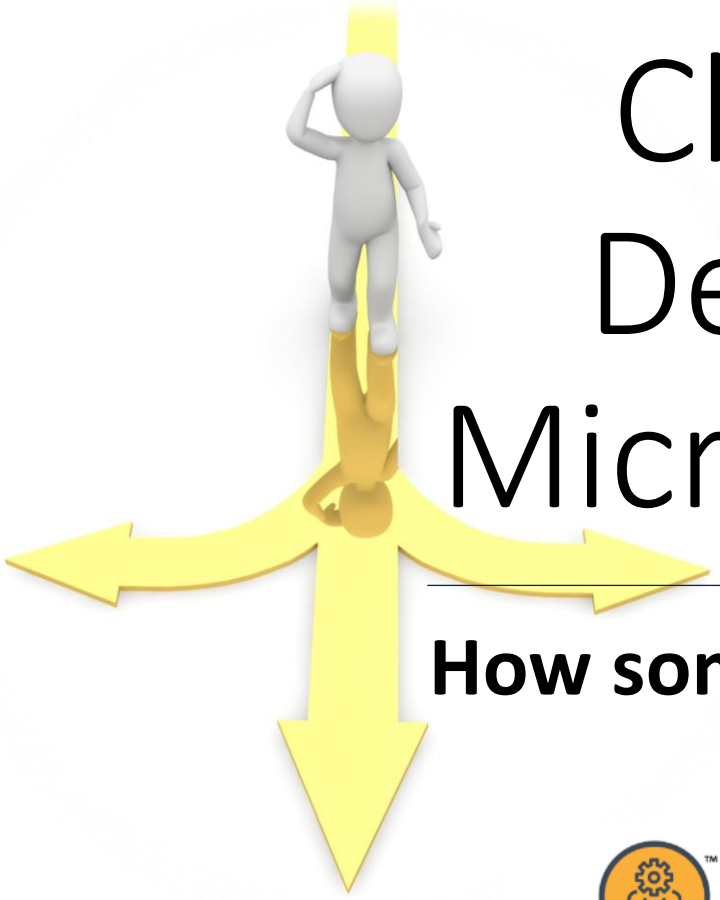

Cloud Confusion, DevOps Dilemma, Microservice Madness

How some enterprises are on cross-roads

DOIS™
An Xellentro Event

DevOps India Summit 2019

#DOIS2019

BMK Lakshminarayanan

@LBMKRISHNA

ABOUT ME

BMK Lakshminarayanan

Solutions Architect | DevOps Advocate

MISSION

Make your life better,

Make others life better,

Make the world better

Community

DEVOPS DAYS

CLOUD NATIVE SUMMIT

CLOUD NATIVE COMPUTING FOUNDATION AMBASSADOR

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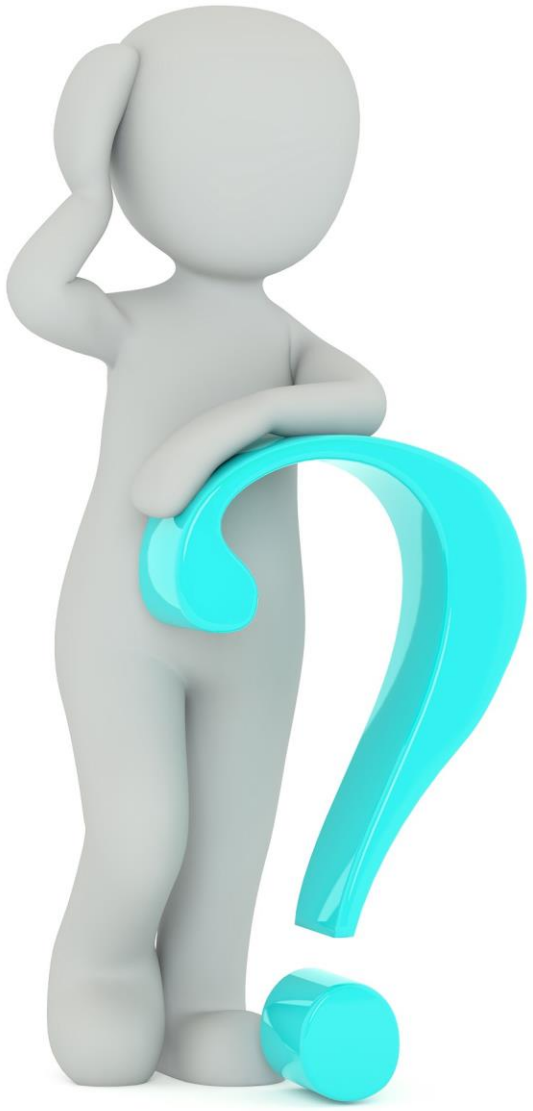
Scene -1

Bob Oopsy, CIO

WDWOD

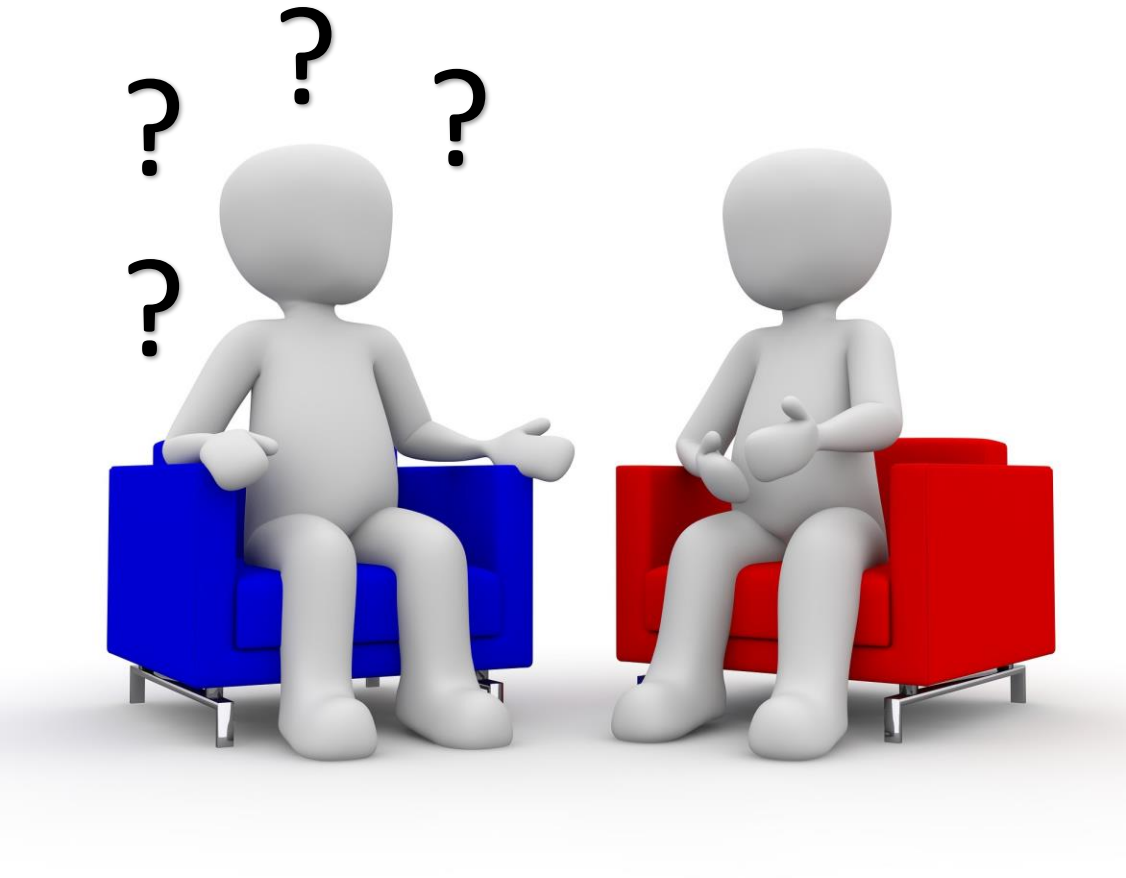


IT Performance

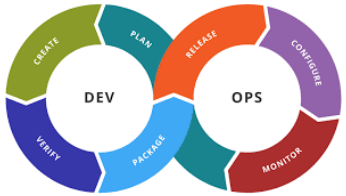
Lower Cost

Speed to Market

Exceed Expectations



Cloud

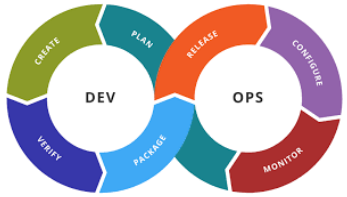


DevOps

Microservices

Scene -2

LEARN


CREATE
PLAN
RELEASE
CONFIGURE
DEV
OPS
VERIFY
PACKAGE
MONITOR

1000 times/day

Scale

A/B Testing

B/G Deployments

Containers

Microservices

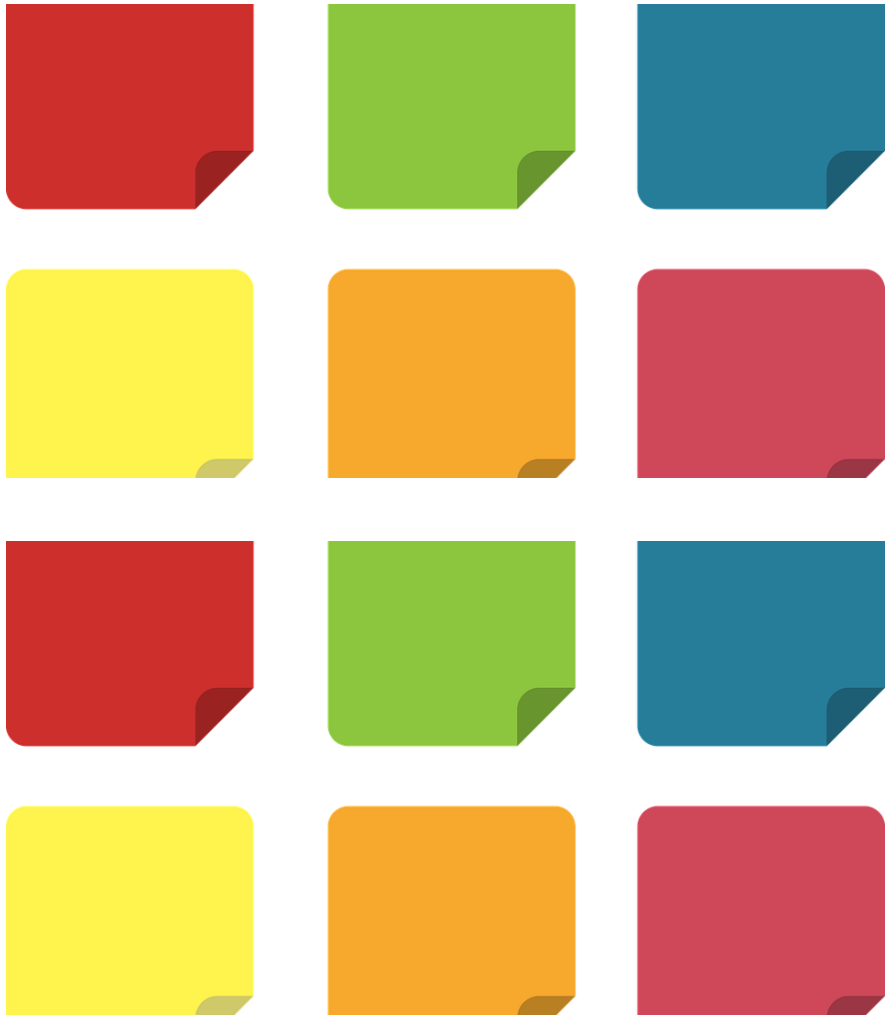
Scene -3

Meeting

ADCKM – Mission 2020

Scene -4

ADCKM – Mission 2020



ADCKM
WDWOD

UPDATE

ADCKM
WDWOD

ADCKM

WPWOP

Scene -5

1st Microservice PROD

Just under 6 weeks

We need more

On our way - Unicorn

Scene -6

Service Outage

Production defect

Honest Status Page
@honest_update

We replaced our monolith with micro services so that every outage could be more like a murder mystery.

4:40 AM · Oct 8, 2015 · Buffer

3K Retweets **2.7K** Likes

Issue investigation

Pointing fingers

Debated post-mortems

SLO, SLI, SLA compromised

Every Head-of protecting own team

Customer Impacts

WDWOD on Headlines

$$$ Impact

Reputational damage

Board is under pressure

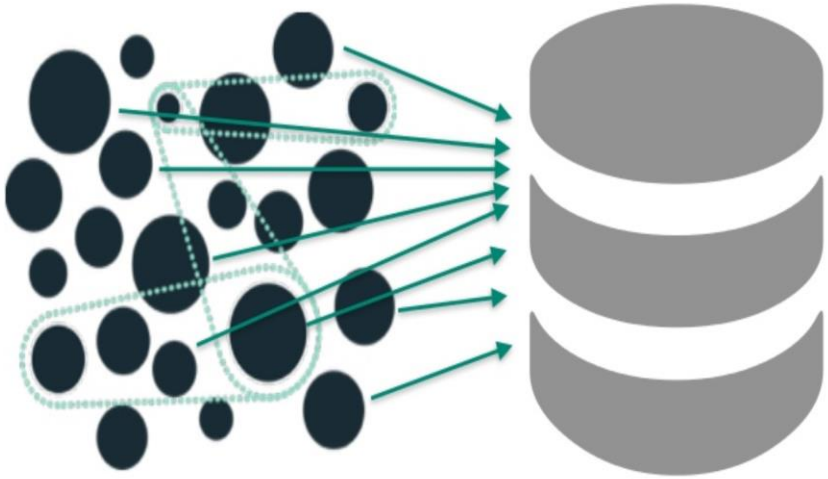
Scene -7

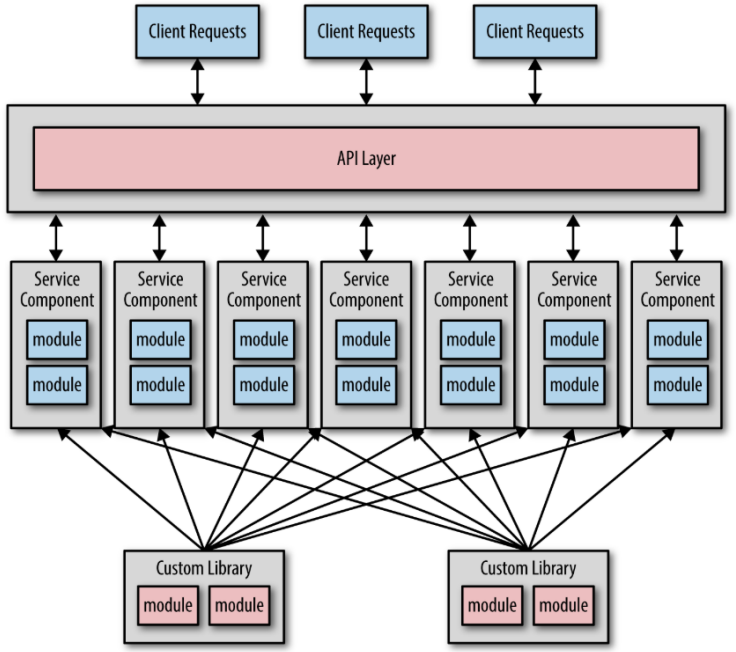
Kelsey Hightower
@kelseyhightower

DevOps doesn't own culture. People do.

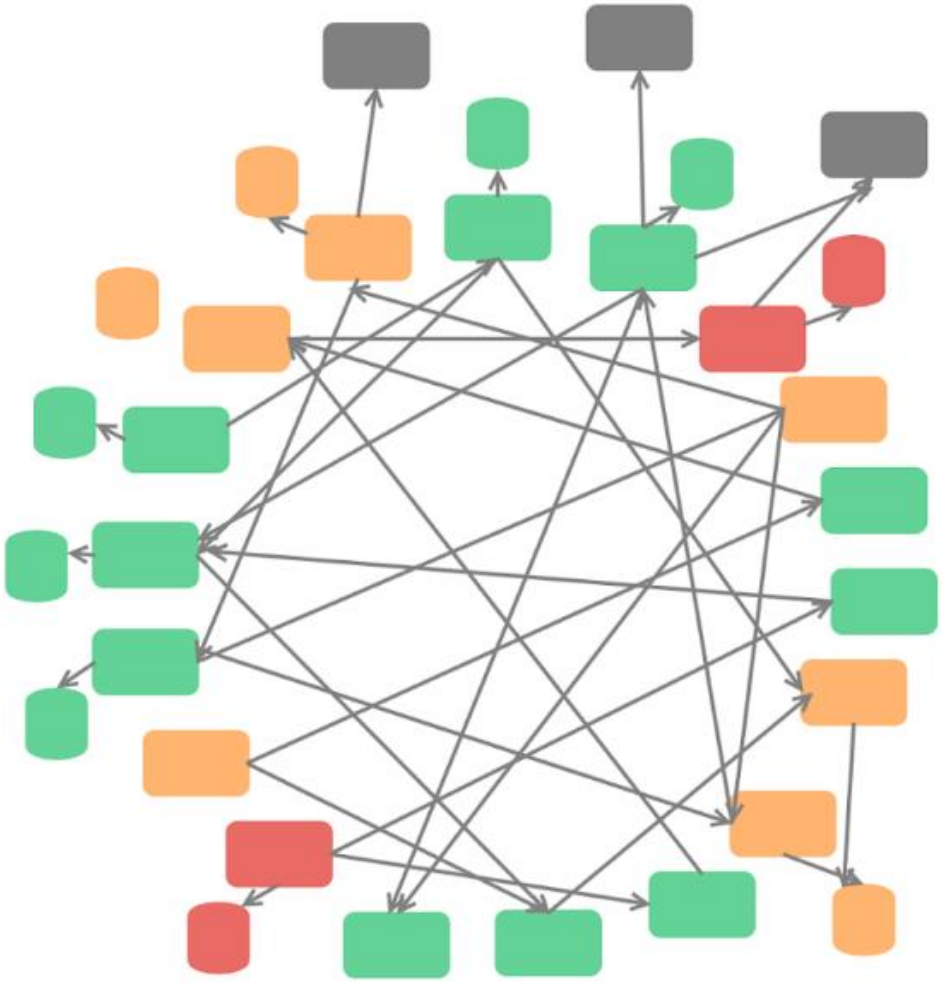
10:52 PM · Aug 31, 2019 · Twitter Web App

39 Retweets **171** Likes





Mum asked me to share things



Service Mess

PaaS DBA Sec Desktop Shared API GW Shared Services

TBD

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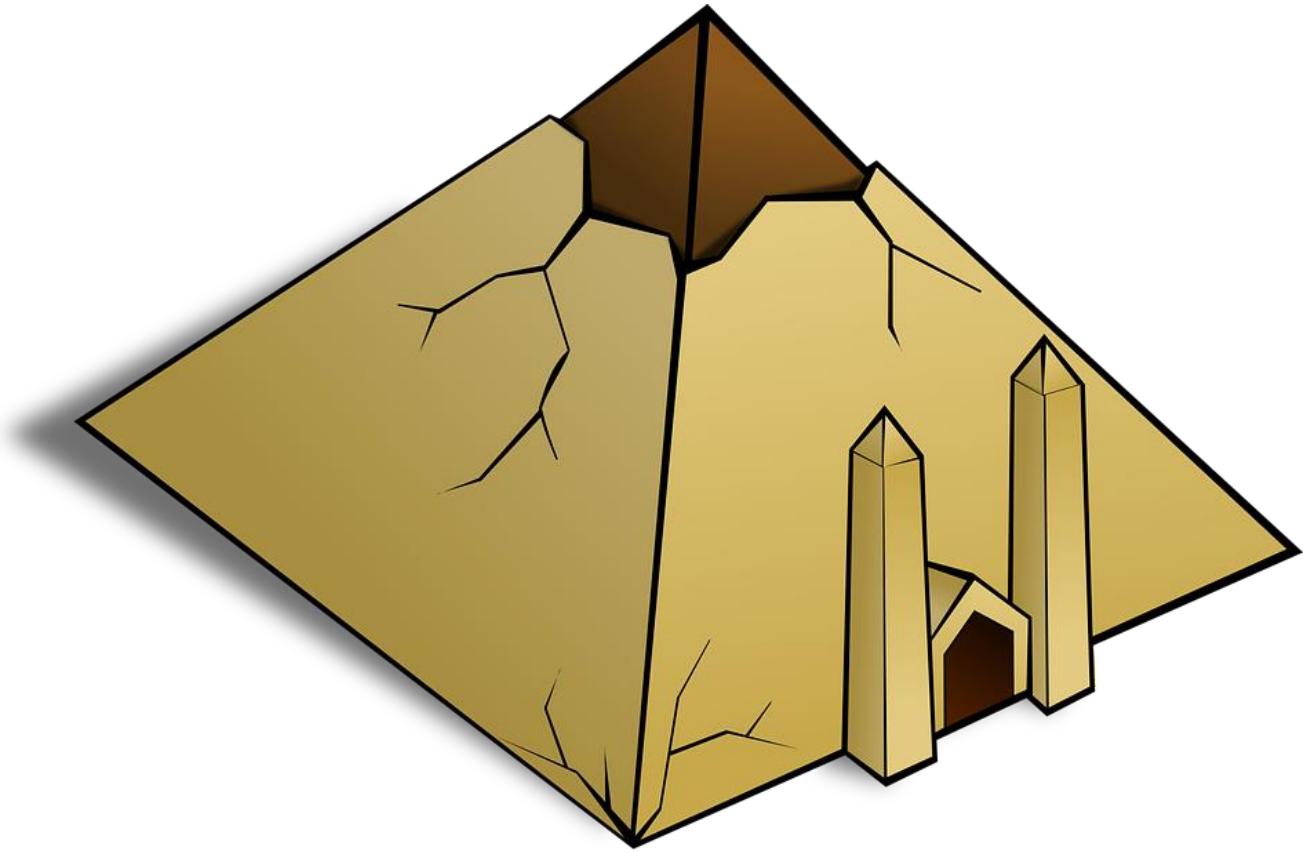

12 factors app and 1 fractured enterprise

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Monoliths better

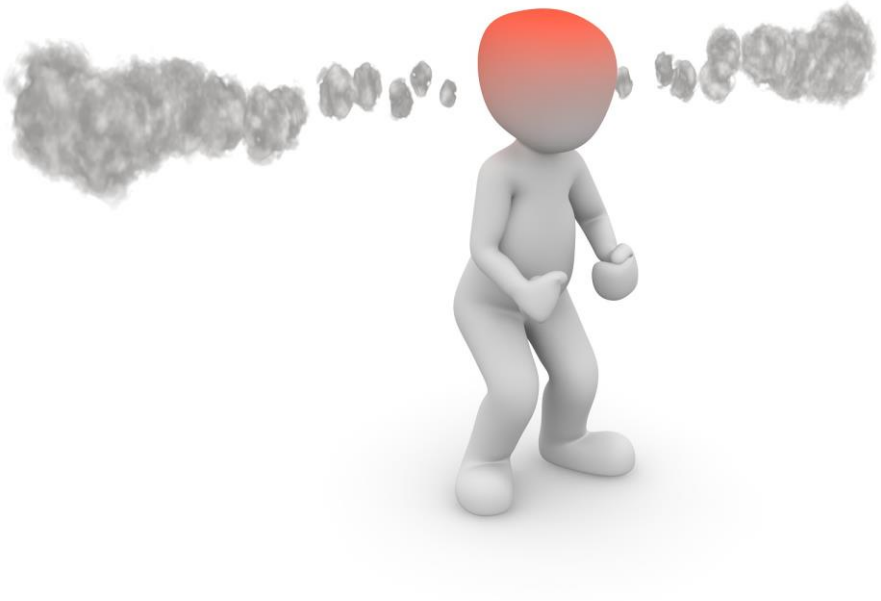
I told you

Not for us

We are not them

Slow is better

Scene -8



Fake IT

Microservices?

Cloud?

DevOps?

A dream
“Stone Soup”

Scene -9

OLD WAYS
WON'T OPEN
NEW DOORS

In banking, you don't often get a clean slate like you would at some of the new tech companies. To transform banking, you not only need to be equipped with the latest technology skills, you also need to transform the culture and skill sets of existing teams, and deal with legacy infrastructure.

—Siew Choo Soh, Managing Director, DBS Bank

Scene -10

“[T]he role of leadership on technology transformation has been one of the more overlooked topics in DevOps, despite the fact that transformational leadership is essential”
—**Accelerate: The Science of Lean Software and DevOps**

The three major kinds of organizational cultures: pathological, bureaucratic, and generative

Pathological	Bureaucratic	Generative
Power-oriented	**Rule-oriented**	**Performance-oriented**
Low cooperation	Modest cooperation	High cooperation
Messengers shot	Messengers neglected	Messengers trained
Responsibilities shirked	Narrow responsibilities	Risks are shared
Bridging discouraged	Bridging tolerated	Bridging encouraged
Failure leads to scapegoating	Failure leads to justice	Failure leads to enquiry
Novelty crushed	Novelty leads to problems	Novelty implemented

Ron Westrum

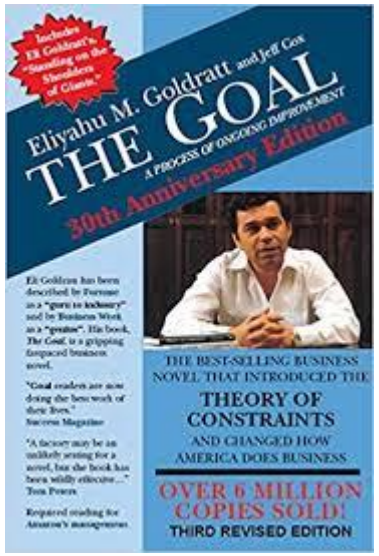
YOU BUILD IT, YOU RUN IT

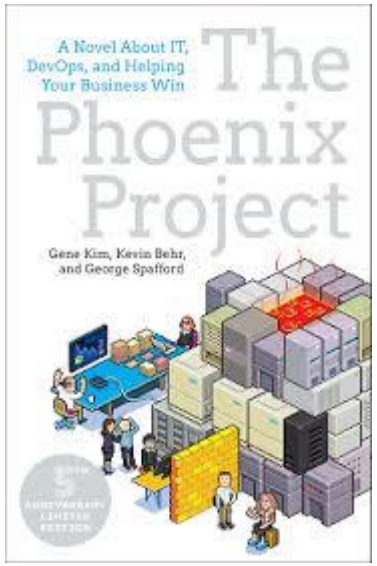
-Werner Voegls

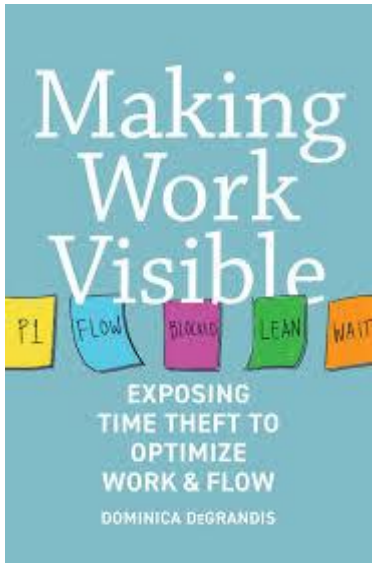
DEVOPS IS NOT A GOAL, BUT A NEVER-ENDING PROCESS OF CONTINUAL IMPROVEMENT

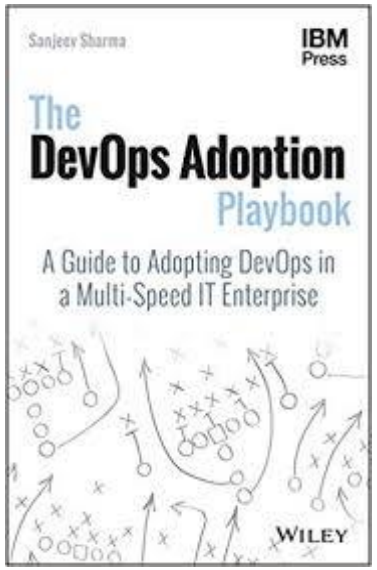
- Jez Humble

It is a journey
Continuous Learning
CALMS

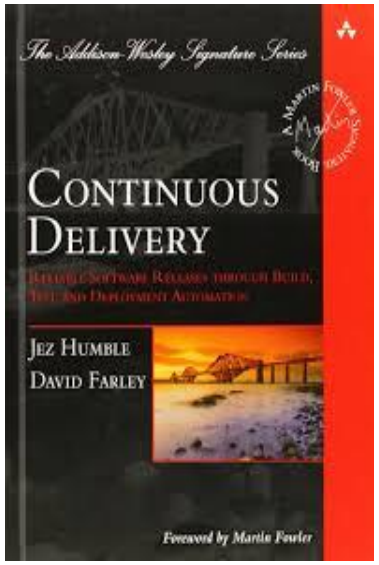

Eliyahu M. Goldratt and Jeff Cox
THE GOAL
A PROCESS OF ONGOING IMPROVEMENT
30th Anniversary Edition
THE BEST-SELLING BUSINESS NOVEL THAT INTRODUCED THE
THEORY OF CONSTRAINTS
AND CHANGED HOW AMERICA DOES BUSINESS
OVER 6 MILLION COPIES SOLD!
THIRD REVISED EDITION

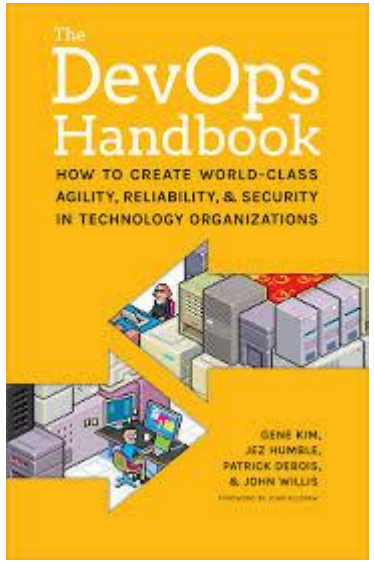

A Novel About IT, DevOps, and Helping Your Business Win
The Phoenix Project
Gene Kim, Kevin Behr, and George Spafford

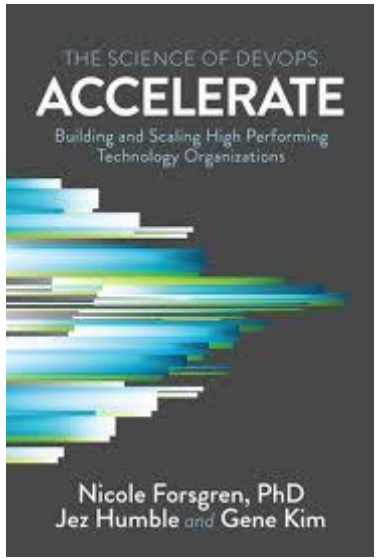

Making Work Visible
EXPOSING TIME THEFT TO OPTIMIZE WORK & FLOW
DOMINICA DeGRANDIS

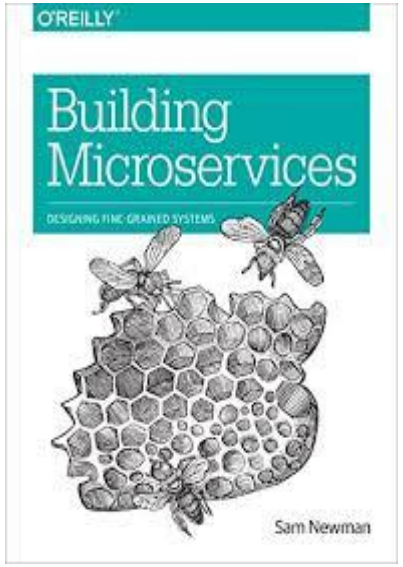

Sanjeev Sharma
IBM Press
The DevOps Adoption Playbook
A Guide to Adopting DevOps in a Multi-Speed IT Enterprise
WILEY

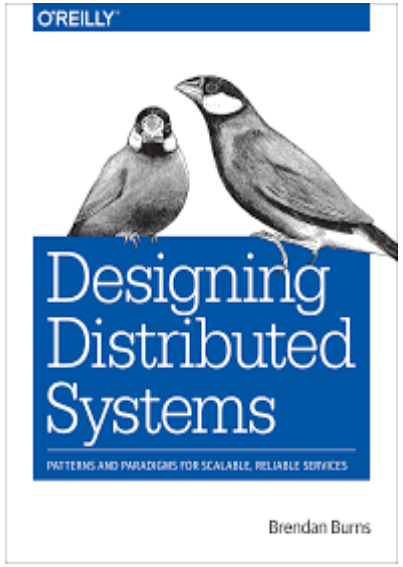
PROJECT TO PRODUCT
MIK KERSTEN
FOREWORD BY GENE KIM

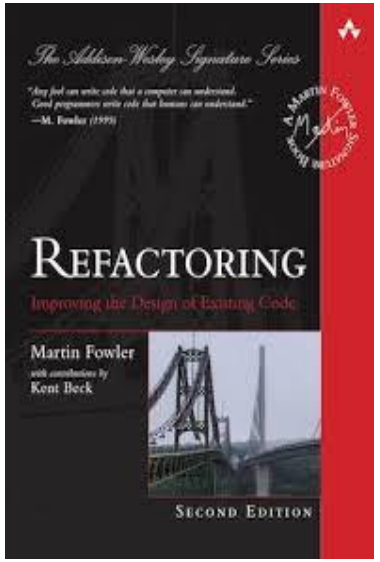

The Addison-Wesley Signature Series
CONTINUOUS DELIVERY
JEZ HUMBLE
DAVID FARLEY
Foreword by Martin Fowler

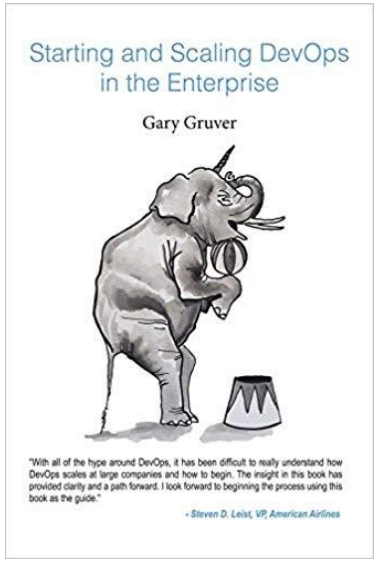

The DevOps Handbook
HOW TO CREATE WORLD-CLASS AGILITY, RELIABILITY, & SECURITY IN TECHNOLOGY ORGANIZATIONS
GENE KIM, JEZ HUMBLE, PATRICK DEBOIS, & JOHN WILLIS

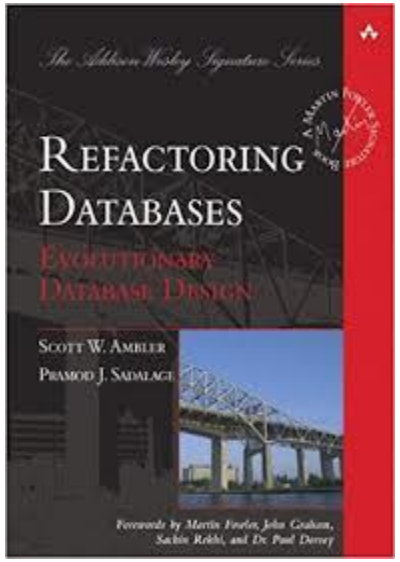

THE SCIENCE OF DEVOPS
ACCELERATE
Building and Scaling High Performing Technology Organizations
Nicole Forsgren, PhD
Jez Humble and Gene Kim

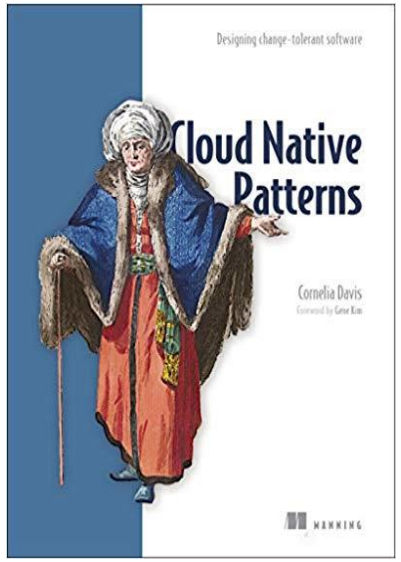

O'REILLY
Building Microservices
DESIGNING FINE-GRAINED SYSTEMS
Sam Newman


O'REILLY
Designing Distributed Systems
PATTERNS AND PARADIGMS FOR SCALABLE, RELIABLE SERVICES
Brendan Burns







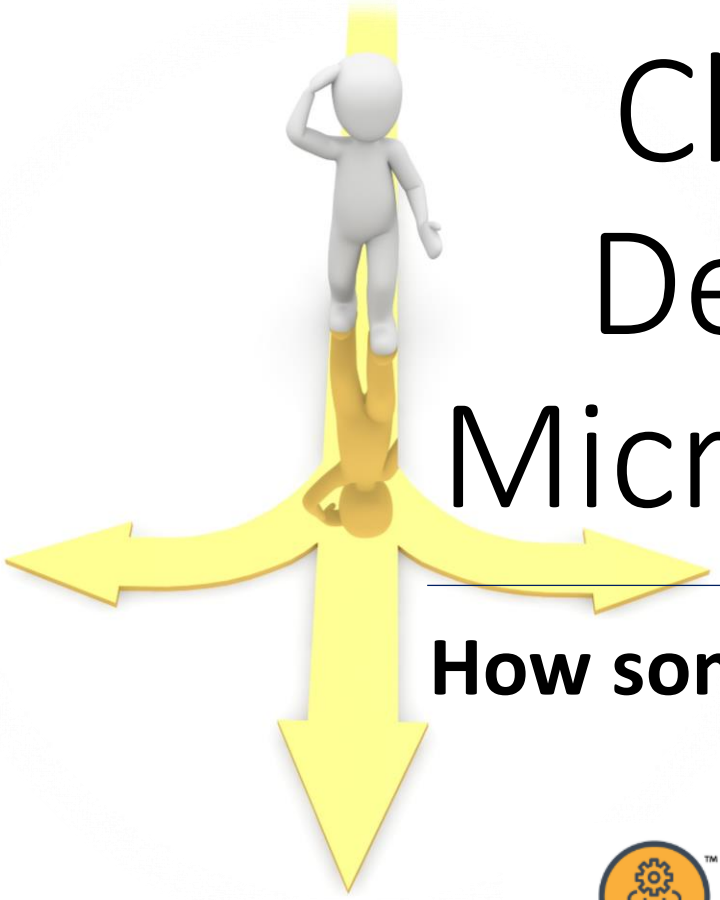


DevOps INSTITUTE

ADVANCING THE HUMANS OF DEVOPS

DORA

DEVOPS RESEARCH & ASSESSMENT



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